

# High-Flow Nasal Cannula Versus Conventional Oxygen Therapy During Bronchoscopy

## A Meta-Analysis of Randomized Controlled Trials

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**Background:** Hypoxemia is a frequent complication during bronchoscopy, and optimal oxygenation strategies remain clinically important. Previous evidence suggests that high-flow nasal cannula (HFNC) may outperform conventional oxygen therapy (COT).

**Methods:** A meta-analysis of randomized controlled trials (RCTs) was conducted by searching PubMed, Embase, and Cochrane databases for studies comparing HFNC with COT in adult patients undergoing flexible bronchoscopy. Outcomes included the incidence of hypoxemia, lowest SpO<sub>2</sub>, procedure duration, patient comfort, and sedation dose. Pooled data were analyzed using a random-effects model, with results reported as risk ratios (RR), mean differences (MD), or standardized mean differences (SMD), as appropriate. Heterogeneity was assessed using the Cochran *Q* test and *I*<sup>2</sup> statistic.

**Results:** Fourteen RCTs (*n* = 2480) were included: 1263 patients (50.9%) received HFNC. HFNC significantly reduced the risk of hypoxemia (RR, 0.41; 95% CI: 0.29–0.59; *P* < 0.0001) and increased minimum SpO<sub>2</sub> (MD, +5.10 percentage points; 95% CI: 3.30–6.91; *P* < 0.0001). There was no significant difference in patient comfort (SMD, –0.07; 95% CI: –0.25 to 0.11; *P* = 0.46) and procedure time (MD, –0.73 min; 95% CI: –1.73 to 0.27; *P* = 0.154). Sedation requirements were slightly higher in the HFNC group (SMD, 0.17; 95% CI: 0.04–0.29; *P* = 0.008).

**Conclusion:** HFNC significantly reduces hypoxemic events and improves oxygenation compared with COT during bronchoscopy. Although anesthetic use slightly increased, there was no difference in procedure time or patient comfort.

**Key Words:** high-flow nasal cannula, conventional oxygen therapy, bronchoscopy, hypoxemia, meta-analysis

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Bronchoscopy remains a fundamental tool in pulmonary medicine, enabling both diagnostic procedures such as transbronchial lung biopsy, endobronchial forceps biopsy, transbronchial needle aspiration (TBNA), and endobronchial ultrasound (EBUS)-guided sampling, and therapeutic interventions including airway stent placement and control of hemoptysis.<sup>1,2</sup>

Despite its clinical utility, bronchoscopy frequently challenges respiratory physiology. The insertion of a bronchoscope narrows the airway lumen, sedation attenuates ventilatory drive, and both can disturb ventilation-perfusion matching and increase airway resistance, predisposing patients to hypoxemia. Individuals with obesity, obstructive sleep apnea (OSA), chronic obstructive pulmonary disease (COPD), interstitial lung disease, or heart failure are at particularly elevated risk of hypoxemia and arrhythmia during bronchoscopy.<sup>3</sup>

Conventional oxygen therapy (COT)—delivered using devices such as nasal cannula, simple face mask, or nonrebreather mask—remains the most frequently used modality during bronchoscopy. However, its effectiveness is limited by variable fractional inspired oxygen (FiO<sub>2</sub>) delivery, lack of positive end-expiratory pressure (PEEP) generation, and inadequate clearance of anatomic dead space.<sup>4</sup> These deficiencies contribute to ongoing risks of hypoxemia, procedural interruption, and potential escalation to invasive ventilation techniques.<sup>5</sup>

High-flow nasal cannula (HFNC) oxygen therapy has emerged as a physiologically attractive alternative. By delivering heated, humidified gas at flow rates that exceed peak inspiratory demand, HFNC provides stable FiO<sub>2</sub>, generates low-level PEEP, reduces nasopharyngeal dead space, and improves mucociliary clearance.<sup>6</sup> Preliminary studies and earlier meta-analyses suggested potential benefits for bronchoscopy, yet were limited by small sample sizes, heterogeneous populations, and incomplete inclusion of recent trials.<sup>7,8</sup>

To address these gaps, we conducted a systematic review and meta-analysis of RCTs evaluating HFNC versus COT in adult patients undergoing bronchoscopy. Our objective was to synthesize the most current evidence on efficacy, safety, and procedural outcomes, thereby informing clinical practice and future guideline development.

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METHODS

This systematic review and meta-analysis adhered to the Cochrane Handbook for Systematic Reviews of Interventions, and reporting followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.<sup>9,10</sup> The protocol was prospectively registered in the International Prospective Register of Systematic Reviews (PROSPERO: CRD420251121239) on May 27, 2025.

Search Strategy and Information Sources

We searched PubMed, Embase, and the Cochrane Central Register of Controlled Trials between June 19 and June 29, 2025, with publication censoring in June 2025. Additional studies were identified through reference screening of prior systematic reviews.

Eligibility Criteria

We included RCTs that: (1) enrolled adults (18 years or older); (2) compared HFNC with COT during flexible bronchoscopy; and (3) explicitly reported HFNC settings (flow rates and FiO<sub>2</sub>) with a conventional oxygen comparator (nasal cannula, face mask, nonrebreather mask, or noninvasive ventilation). Exclusion criteria were: pediatric populations; patients already invasively ventilated; studies

evaluating HFNC only for preoxygenation or postprocedure recovery; patients undergoing rigid bronchoscopy; or trials conducted exclusively in acute hypoxemic respiratory failure.

Study Selection and Data Extraction

Two independent reviewers screened titles, abstracts, and full texts, with discrepancies resolved by a third reviewer. Data extraction included trial characteristics, patient demographics, procedural details, and outcomes (Fig. 1).

Outcomes

The primary outcome was the incidence of desaturation events. Secondary outcomes included the lowest SpO<sub>2</sub>, procedure time, patient comfort, and sedation dose.

Risk of Bias Assessment

The methodological quality of the included randomized controlled trials was evaluated using the Cochrane Collaboration's Risk of Bias 2 (RoB-2).<sup>11</sup> This instrument assesses 5 key domains: selection bias, performance bias, detection bias, attrition bias, and reporting bias. Each study was subsequently categorized as having a low, high, or unclear risk of bias.

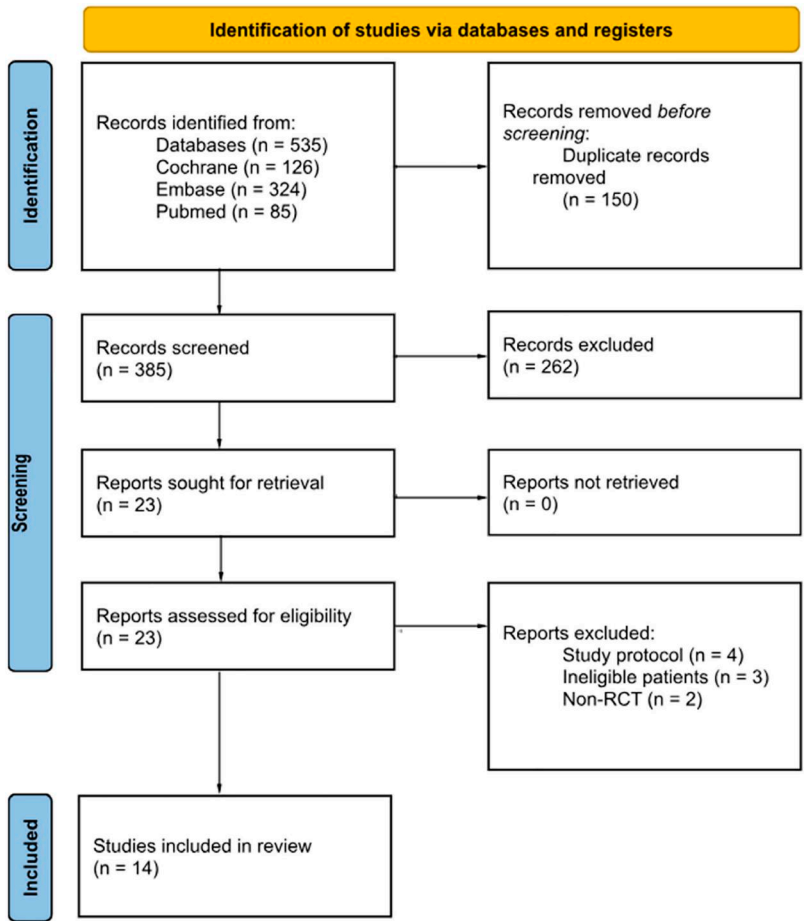


FIGURE 1. The flowchart of the study selection process. PRISMA flow diagram of study screening and selection. PRISMA indicates Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

TABLE 1. Baseline Patient and Study Characteristics

| References                       | Procedure type          | Control ventilation         | Sample size |     | Baseline SpO <sub>2</sub> (%) |            | Age (y)     |             | Males (n) |     | BMI (kg/m <sup>2</sup> )* |            |
|----------------------------------|-------------------------|-----------------------------|-------------|-----|-------------------------------|------------|-------------|-------------|-----------|-----|---------------------------|------------|
|                                  |                         |                             | HFNC        | COT | HFNC                          | COT        | HFNC        | COT         | HFNC      | COT | HFNC                      | COT        |
| Burton et al <sup>14</sup>       | EBUS/bronchoscopy       | Conventional nasal cannula  | 60          | 61  | 94.0 ± 3.0                    | 93.0 ± 3.0 | 67.0 ± 14.0 | 61.0 ± 15.0 | 32        | 33  | 28.0 ± 7.4                | 27.9 ± 5.9 |
| Daric et al <sup>15</sup>        | Flexible bronchoscopy   | Conventional nasal cannula  | 295         | 305 | NA                            | NA         | 69.0 ± 10.4 | 68.7 ± 10.4 | 193       | 210 | 24.1 ± 4.8                | 25.2 ± 5.6 |
| Douglas et al <sup>16</sup>      | EBUS                    | Bite block                  | 30          | 30  | 96.7 ± 3.0                    | 96.0 ± 3.0 | 62.8 ± 14.1 | 63.4 ± 14.3 | 19        | 19  | 26.9 ± 5.8                | 27.4 ± 6.4 |
| Gavrieli et al <sup>17</sup>     | EBUS                    | Conventional nasal cannula  | 20          | 40  | 98.4 ± 1.8                    | 99.4 ± 1.7 | 63.5 ± 14.8 | 66.4 ± 10.8 | 15        | 23  | 25.5 ± 4.5                | 26.7 ± 4.3 |
| Gupta et al <sup>3</sup>         | EBUS                    | Conventional nasal cannula  | 61          | 61  | 94.9 ± 2.2                    | 96.4 ± 1.8 | 50.5 ± 17.0 | 47.5 ± 15.2 | 36        | 32  | 24.4 ± 3.4                | 24.1 ± 3.4 |
| Irfan et al <sup>18</sup>        | EBUS                    | Nasal prongs                | 20          | 20  | 98.4 ± 2.7                    | 96.9 ± 1.9 | 61.9 ± 12.0 | 64.5 ± 14.0 | 12        | 12  | NA                        | NA         |
| Longhini et al <sup>19</sup>     | Fibreoptic bronchoscopy | Conventional oxygen therapy | 18          | 18  | NA                            | NA         | NA          | NA          | 15        | 12  | 26.2 ± 5.7                | 25.1 ± 4.2 |
| Lucangelo et al <sup>20</sup>    | Fibreoptic bronchoscopy | Venturi mask                | 15          | 15  | 93.9 ± 4.0                    | 94.3 ± 2.4 | 68.9 ± 12.2 | 69.4 ± 13   | 8         | 9   | 24.7 ± 5.3                | 26 ± 5.3   |
| Ben-Menachem et al <sup>21</sup> | Flexible bronchoscopy   | Conventional nasal cannula  | 37          | 39  | 98.0 ± 1.5                    | 98.0 ± 1.5 | 54.9 ± 11.7 | 55 ± 11.9   | 22        | 29  | 25.2 ± 4.1                | 24.5 ± 4.1 |
| Qin et al <sup>22</sup>          | Flexible bronchoscopy   | Conventional nasal cannula  | 72          | 70  | 97.9 ± 2.8                    | 97.4 ± 3.2 | 54.9 ± 23.8 | 51.7 ± 22.1 | 37        | 37  | 21.2 ± 3.1                | 22.2 ± 3.7 |
| Sharma et al <sup>23</sup>       | EBUS                    | Conventional nasal cannula  | 30          | 30  | 97.2 ± 1.9                    | 97.5 ± 2.0 | 62.5 ± 7.4  | 59.9 ± 8.7  | 24        | 26  | 20.4 ± 2.6                | 20.2 ± 2.7 |
| Ucar et al <sup>24</sup>         | EBUS/bronchoscopy       | Conventional nasal cannula  | 85          | 85  | 94.1 ± 2.0                    | 94.3 ± 2.0 | 57.8 ± 14.0 | 57.5 ± 14.0 | 66        | 65  | 27.3 ± 3.0                | 26.4 ± 4.0 |
| Wang et al <sup>25</sup>         | Flexible bronchoscopy   | Nasal prongs                | 392         | 396 | 95.6 ± 2.2                    | 95.6 ± 2.2 | 58.0 ± 11.9 | 57.5 ± 13.3 | 204       | 222 | 23.6 ± 3.4                | 23.5 ± 3.7 |
| Zhang et al <sup>26</sup>        | EBUS                    | Facial mask                 | 87          | 89  | 98.6 ± 2.2                    | 99.0 ± 1.5 | 64.2 ± 9.3  | 63.6 ± 7.7  | 13        | 16  | 23.6 ± 2.8                | 23.8 ± 2.9 |

\*Mean or median.

BMI indicates body mass index; COT, conventional oxygen therapy; EBUS, endobronchial ultrasound; HFNC, high-flow nasal cannula; NA, not available.

## Effect Measures and Data Synthesis

For dichotomous outcomes (desaturation events), risk ratios (RR) with 95% CI were calculated. For continuous outcomes, mean differences (MD) were computed for the lowest SpO<sub>2</sub> and procedure duration, whereas standardized mean differences (SMD) were applied for patient comfort and sedation dose. Pooled effect estimates were generated using the inverse variance method.

Statistical heterogeneity was assessed with the Cochran *Q* test and quantified using the *I*<sup>2</sup> statistic, with *I*<sup>2</sup> ≥ 25% considered indicative of significant heterogeneity. Anticipating variability in study populations, procedural techniques, and interventions, a random-effects model with the restricted maximum-likelihood estimator for  $\tau^2$  was applied to all outcomes.

When trials reported medians and interquartile ranges rather than means and SDs, these values were converted using the method described by Wan and Luo, as recommended by the Cochrane Handbook.<sup>9,12</sup> All statistical analyses were performed with R software, version 4.4.1 (R Foundation for Statistical Computing, Vienna, Austria). Missing means and SDs were estimated from medians and interquartile ranges using the method proposed by Wan and Luo, as per Cochrane recommendations. In addition, we performed sensitivity analyses using a leave-one-out approach.

## Certainty of Evidence

The certainty of evidence was graded according to the GRADE framework, which evaluates 6 domains: study design, risk of bias, inconsistency, indirectness, imprecision, and other considerations. Evidence for each outcome was classified as high, moderate, low, or very low certainty. A summary of findings table was generated using GRADEpro software (version 3.6.1).<sup>13</sup>

## RESULTS

### Study Selection and Characteristics

Our initial search strategy identified 535 articles, of which 150 were duplicates. After screening, 23 studies were selected for full-text review. Applying the predefined eligibility criteria, 14 studies met the inclusion criteria and were included in the analysis. All included studies were randomized controlled trials (RCTs), encompassing a total of 2480 patients, of whom 1263 (50.9%) received high-flow nasal cannula (HFNC). The procedures evaluated across the included trials were nasal fibroscopy, linear and radial EBUS, flexible bronchoscopy, transbronchial needle aspiration, standard bronchoscopy, and nasal bronchoscopy. Baseline characteristics of the study populations are summarized in Table 1.

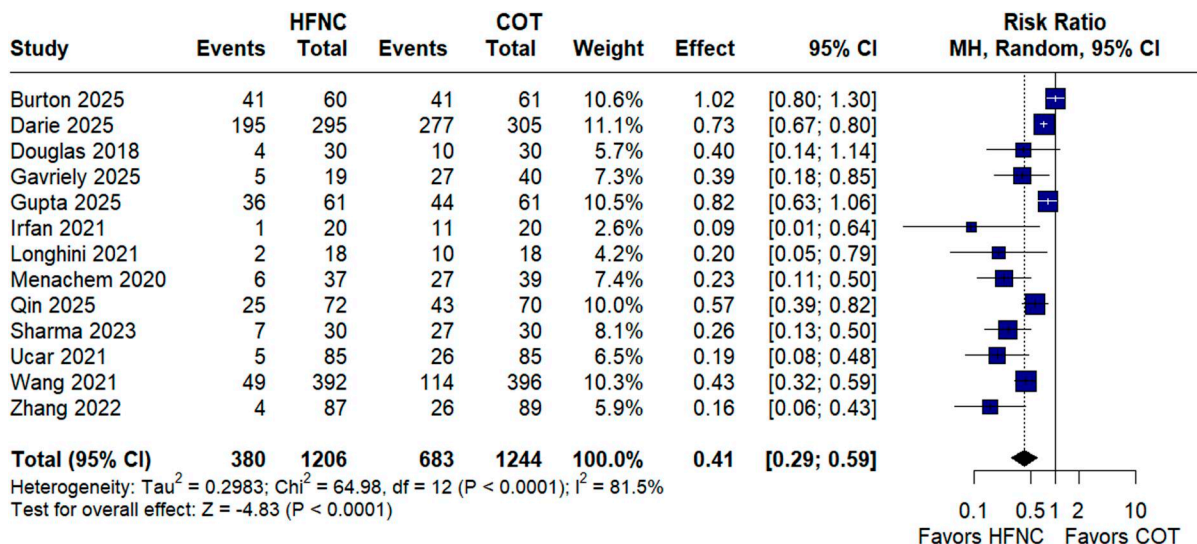
### Pooled Analysis of All Studies

#### Desaturation Events

The incidence of desaturation events was significantly lower in the HFNC group compared with the COT group (RR, 0.41; 95% CI: 0.29-0.59; *P* < 0.0001; *I*<sup>2</sup> = 81.5%; 13 RCTs; 2450 patients; Fig. 2).

#### Lowest SpO<sub>2</sub>

The lowest SpO<sub>2</sub> values were significantly higher in the HFNC group compared with the COT group (MD, 5.10;



**FIGURE 2.** Desaturation events. HFNC significantly reduced the incidence of desaturation events compared with COT, indicating improved oxygenation and greater procedural safety.

95% CI: 3.30-6.91;  $P < 0.0001$ ;  $I^2 = 87.4\%$ ; 10 RCTs; 2099 patients; Fig. 3).

### Procedure Time

Procedure duration did not differ significantly between patients in the HFNC group compared with the COT group (MD,  $-0.73$  min; 95% CI:  $-1.73$  to  $0.27$ ;  $P = 0.1546$ ;  $I^2 = 51.1\%$ ; 10 RCTs; 2112 patients; Fig. 4).

### Patient Comfort

No significant difference in patient comfort was observed between the HFNC and COT groups (SMD,  $-0.09$ ; 95% CI:  $-0.29$  to  $0.12$ ;  $P = 0.4016$ ;  $I^2 = 50.3\%$ ; 8 RCTs; 1186 patients; Fig. 5).

### Sedation Dose

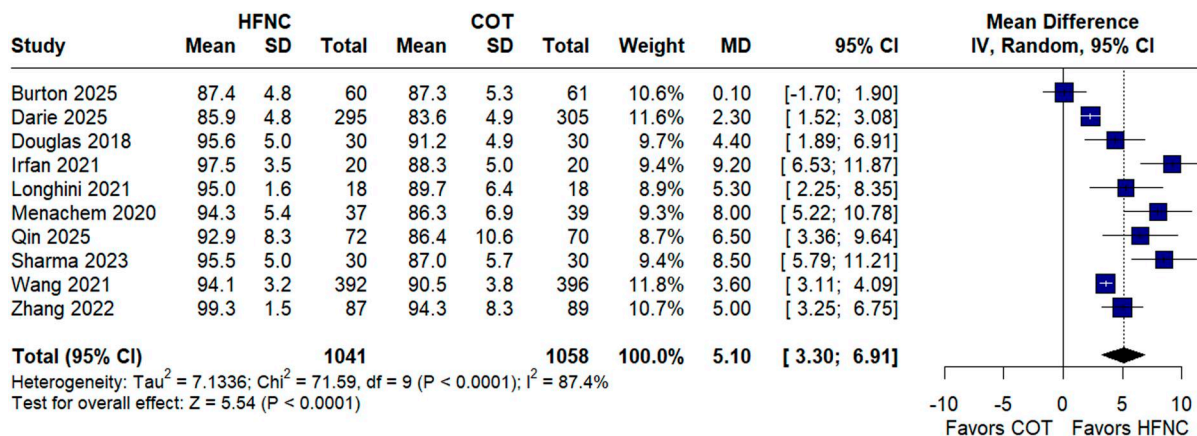
The required sedation dose was significantly lower in the COT group compared with the HFNC group (SMD,  $0.18$ ; 95% CI:  $0.05$ - $0.31$ ;  $P = 0.0059$ ;  $I^2 = 0.0\%$ ; 5 RCTs; 945 patients; Supplemental Digital Content 1, <http://links.lww.com/LBR/A402>).

### Sensitivity Analysis

The results for desaturation events, lowest  $SpO_2$ , and patient satisfaction remained consistent in leave-one-out sensitivity analyses, indicating that no single study disproportionately influenced the pooled effect estimates (Supplemental Digital Contents 4, 5, and 6, <http://links.lww.com/LBR/A402>).

Excluding Zhang et al<sup>26</sup> resulted in a statistically significant reduction in procedure time, favoring the HFNC

## Lowest SpO<sub>2</sub>



**FIGURE 3.** Lowest  $SpO_2$ . HFNC significantly increased the lowest  $SpO_2$  values compared with COT, indicating better oxygenation during the procedure.



## Procedure Time

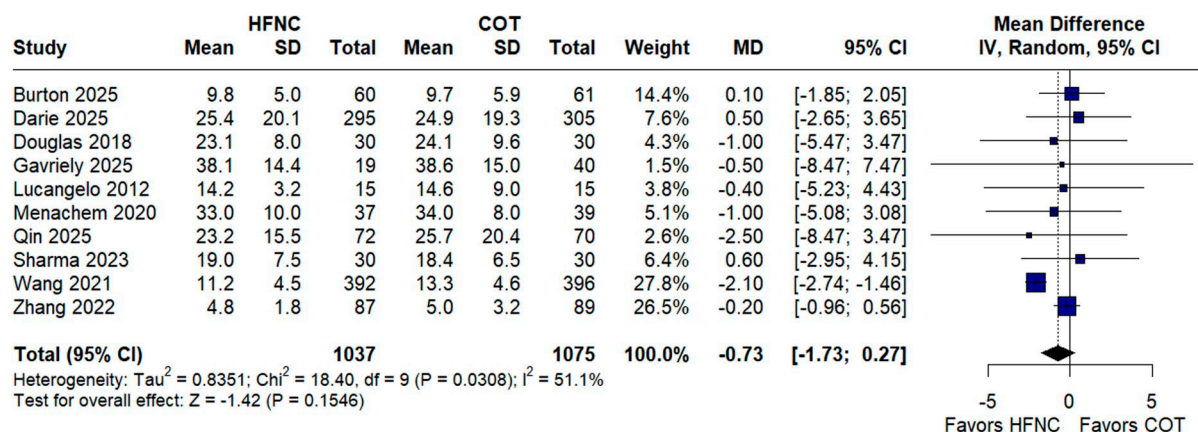


FIGURE 4. Procedure time. HFNC showed no significant difference in procedure time compared with COT.

group compared with the COT group (MD, -1.29; 95% CI: -2.14 to -0.44; 9 RCTs; 1937 patients; Supplemental Digital Content 7, <http://links.lww.com/LBR/A402>).

For the sedation dose, exclusion of Darie et al<sup>15</sup> resulted in the loss of statistical significance (SMD, 0.12; 95% CI: -0.10 to 0.33; 4 RCTs; 0% heterogeneity; Supplemental Digital Content 8, <http://links.lww.com/LBR/A402>), suggesting a moderate influence of this trial on the overall estimate.

### Risk of Bias and Certainty of Evidence

Most trials were judged to be at low risk of bias, with concerns primarily related to lack of blinding. No study was rated high risk (Supplemental Digital Content 2, <http://links.lww.com/LBR/A402>). Certainty of evidence was moderate for desaturation events, lowest SpO<sub>2</sub>, and sedation dose; high for procedure duration; and low for patient comfort (Supplemental Digital Content 3, <http://links.lww.com/LBR/A402>).

### DISCUSSION

In this meta-analysis of 14 RCTs including 2480 adults undergoing bronchoscopy, HFNC significantly reduced the incidence of hypoxemia by 60% compared with

conventional oxygen therapy (RR, 0.41; 95% CI: 0.29-0.59;  $P < 0.0001$ ). HFNC also increased the minimum SpO<sub>2</sub> by an average of 5.1 percentage points. No significant difference was observed in patient comfort and procedure time, whereas sedation requirements were slightly higher with HFNC. These results were consistent across diagnostic and therapeutic bronchoscopies, including high-risk subgroups such as OSA, COPD, and lung transplant recipients.

While earlier meta-analyses concluded that HFNC improves oxygenation and reduces desaturation events compared with low-flow oxygen, our synthesis expands this evidence by incorporating 8 recent RCTs, broadening procedural coverage to include rigid bronchoscopy under general anesthesia and encompassing large high-risk cohorts.<sup>27</sup> We also evaluated additional outcomes not consistently evaluated in previous reviews—procedure duration, sedation dose, and comfort—and included direct comparison HFNC versus noninvasive ventilation (NIV). Although earlier reviews suggested equivalence between HFNC and NIV for oxygenation, a recent RCT demonstrated that, while NIV may achieve slightly higher PaO<sub>2</sub>, HFNC offers superior tolerability and procedural access.<sup>23</sup>

Consistent trial-level benefits underpin the pooled results. Wang et al halved desaturation rates in a large

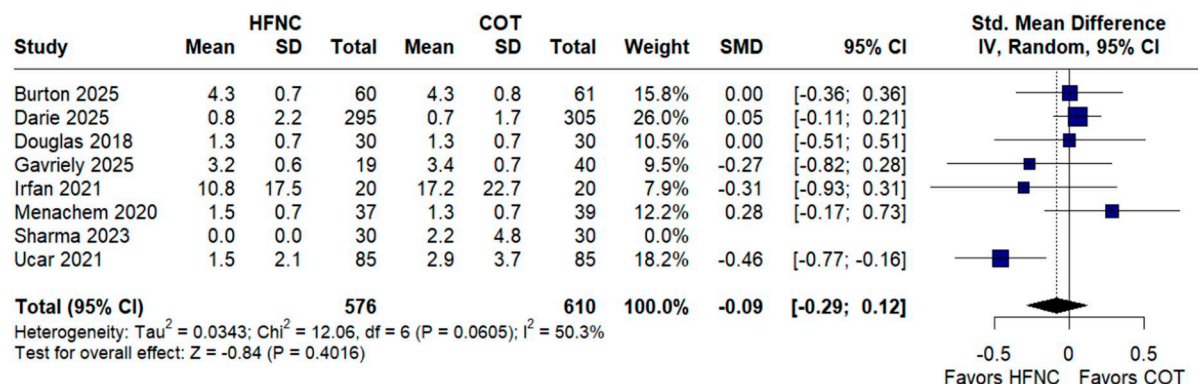


FIGURE 5. Patient comfort. HFNC showed no significant difference in patient comfort compared with COT, indicating comparable tolerability between the 2 methods.

general cohort ( $n=788$ ).<sup>25</sup> Darie et al<sup>17</sup> reported a 53% reduction in cumulative hypoxemia time.<sup>15</sup> Zhang et al reported a >6-fold reduction in hypoxemia in OSA-risk patients.<sup>26</sup> Ben-Menachem et al demonstrated substantial benefit in lung transplant recipients, reducing desaturation events and procedure interruptions.<sup>21</sup> Pikman Gavriely et al found HFNC comparable to laryngeal mask airway (LMA) and superior to nasal cannula in advanced EBUS.<sup>28</sup> Gupta et al observed a significant benefit in patients with cardiopulmonary comorbidities without increased risk of hypercapnia.<sup>3</sup> By contrast, Burton et al reported no oxygenation advantage, likely due to strict  $\text{FiO}_2$  matching, lighter sedation, and lower-risk patients.<sup>14</sup>

HFNC benefits derive from delivery of stable high  $\text{FiO}_2$  at flows exceeding inspiratory demand, generation of low-level positive airway pressure, dead space washout enhancing  $\text{CO}_2$  clearance, and heated humidification improving secretion clearance and comfort.<sup>29,30</sup> Across trials, HFNC rarely worsened  $\text{CO}_2$  retention.<sup>3</sup>

Slightly higher sedation doses with HFNC, seen in our analysis and in Sharma et al may reflect clinicians' confidence in a greater oxygen reserve, allowing deeper sedation. Comfort ratings were generally similar or better with HFNC compared with COT and superior to NIV, which can impede procedural access.<sup>23</sup>

Importantly, leave-one-out sensitivity analyses indicated that excluding Darie et al<sup>15</sup> rendered the sedation dose nonsignificant, and exclusion of Zhang et al<sup>26</sup> materially changed the procedure time estimate, highlighting the sensitivity of these secondary outcomes to individual trials.

Residual heterogeneity likely stems from variable hypoxemia definitions, differences in procedure type and sedation depth, and oxygen delivery settings (flow rates,  $\text{FiO}_2$  matching). As in all meta-analyses, variation in outcome definitions, particularly hypoxemia thresholds and measurement methods, may limit comparability across trials and contribute to heterogeneity.

These findings support HFNC as the preferred oxygenation strategy for bronchoscopy when minimizing desaturation is a priority, particularly in high-risk populations. While NIV can provide greater oxygenation in select contexts, HFNC's balance of efficacy, comfort, and procedural feasibility favors its broader use.

Strengths of this review include protocol registration, a comprehensive and current search strategy, inclusion of only RCTs, and examination of both physiological and patient-centered outcomes. Limitations include variability in definitions and reporting, lack of blinding, and underreporting of long-term clinical outcomes. In addition, secondary outcomes (procedure time, sedation dose, and patient comfort) were not consistently reported, reducing the number of trials included in pooled analyses and potentially introducing reporting bias.

Future research should compare HFNC directly with NIV in pragmatic multicenter trials, establish standardized definitions of hypoxemia, and explore cost-effectiveness and workflow integration.

## CONCLUSIONS

In this meta-analysis of 14 RCTs ( $n=2480$ ), HFNC significantly improved oxygenation during bronchoscopy, reducing desaturation events and increasing nadir  $\text{SpO}_2$ , while preserving procedure duration and patient comfort.

Although sedation requirements were slightly higher, HFNC demonstrated an overall favorable profile of efficacy, safety, and procedural feasibility, supporting its use as a frontline oxygenation strategy in bronchoscopy.

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